

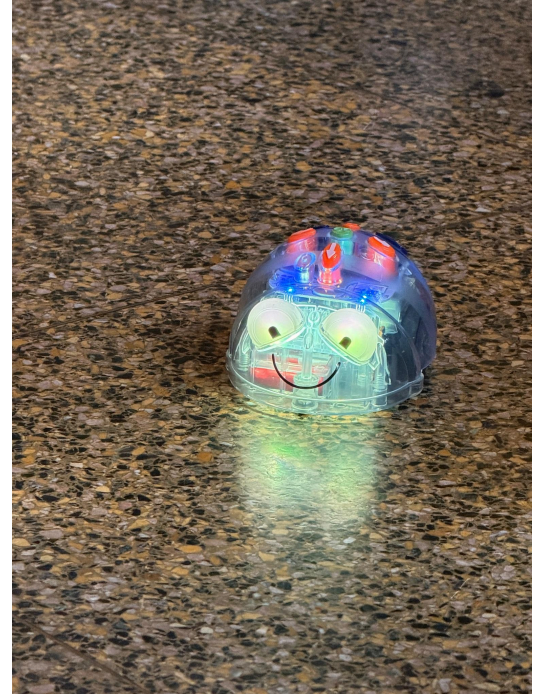
Bee-Bot Garden Prototype

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Overview & Purpose

The Bee-Bot Garden Prototype is the **most fully developed Wellbotics system**, combining **physical computing, game-based learning, and social-emotional learning (SEL)** into a single, facilitated experience for young learners. Developed as part of the *Wellbotics Game Suite Proposal* in partnership with Harlem CoLab LLC, the prototype is explicitly designed to help children build **lifelong emotional and cognitive skills** by drawing an analogy between **programming a robot** and **recognizing and managing one's own emotions and behaviors**.

While the primary goal is to positively impact learners aged **5–8 (K–2)**, a secondary and intentional outcome is to **strengthen facilitators' capacity** to support SEL through play, reflection, and intentional game design.



Target Learners & Context

- **Age Group:** 5–8 years old (K–2)
- **Learning Environment:** Public school settings
- **Contextual Focus:**
 - Schools serving predominantly Black and Hispanic communities
 - Learners facing systemic challenges (economic instability, limited access to resources)
 - Inclusive of neurodiverse students
- **Example Context:** PS241, The STEM School (Harlem, District 3)

The game is intentionally designed to be **tactile, kinesthetic, open-ended, and emotionally supportive**, aligning with the needs of neurodiverse learners and students who benefit from embodied and collaborative learning.

User Story

As a group of no more than four 5- to 8-year-old learners in a public school environment, we want to be engaged in a game where we program a robot with facilitator support to solve engaging problems, so that we can clearly see how

adjusting the robot's programmatic behaviors through sequencing and debugging (NYS CS Standard K1.CT.6) mirrors recognizing and managing our own emotions and behaviors (NYS SEL Benchmark 1A.3A), helping us build lifelong social and emotional skills.

The user story ensures alignment between learner identity, learning goals, emotional outcomes, and facilitation needs.

Emotional Target

Primary Emotional Target: “Enlightened”

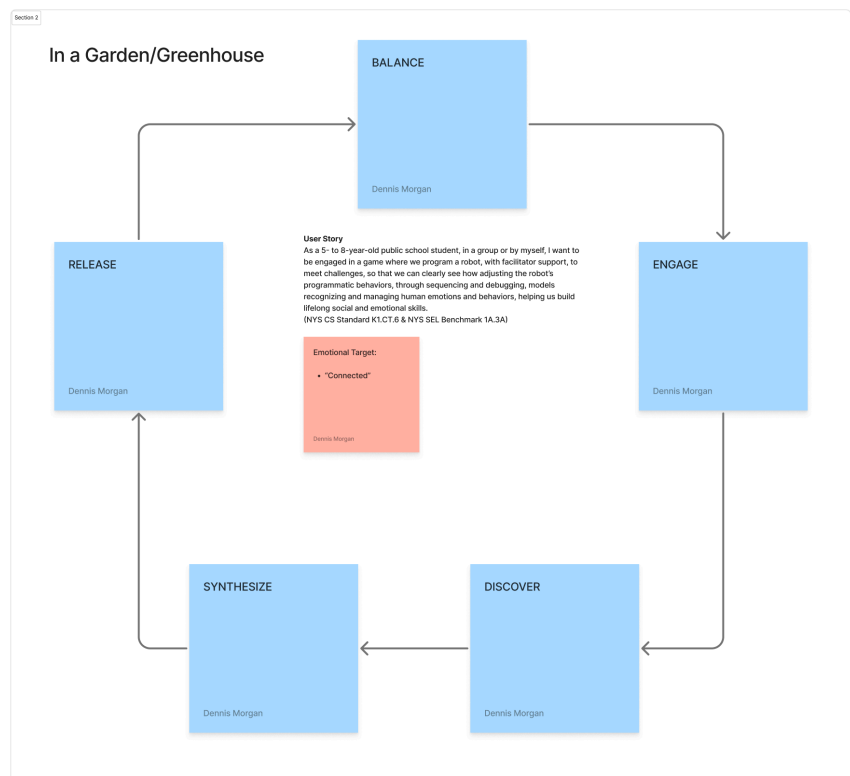
Learners are intended to feel:

- A sense of understanding
- Awareness of cause and effect (both robotic and emotional)
- Connection between action, outcome, and reflection

Core Learning Loop

The Bee-Bot prototype operationalizes the Wellbotics learning loop:

1. **BALANCE** – Initial imbalance or challenge
2. **ENGAGE** – Active exploration and solution testing
3. **DISCOVER** – Clear feedback and “aha” moments
4. **SYNTHESIZE** – Reflection and personal connection
5. **RELEASE** – Meaningful reset and readiness for the next challenge
→ returns to **BALANCE**



Reflection and emotional processing are **structural phases of play**, not post-hoc discussion.

World & Narrative Structure: The Garden Ecosystem

Setting

- **World:** Garden / Greenhouse ecosystem
- **Metaphor:** Insects living and working within a shared environment

Roles

- **Robot Role:** Insect (Bee-Bot represents an insect)
- **Player Role:** Programmer of the insect (robot)
- **Player Avatar:** Insect
 - The player is explicitly programming *a version of themselves*
 - Learners are encouraged to project emotions, intentions, and decisions onto the robot
- **Facilitator Role (Queen Bee):**
 - Functions as *the Earth* or environment
 - Controls contextual conditions (e.g., photosynthesis, “sunny day”)
 - Acts as a **Dungeon-Master-like guide**, embedding learning objectives into the world through artifacts rather than direct instruction
 - Visually represents Hive Health (e.g., costume changes, gestures, props)

Game Board & Spatial Design

Garden Tiles

The board is divided into tiles representing different garden conditions:

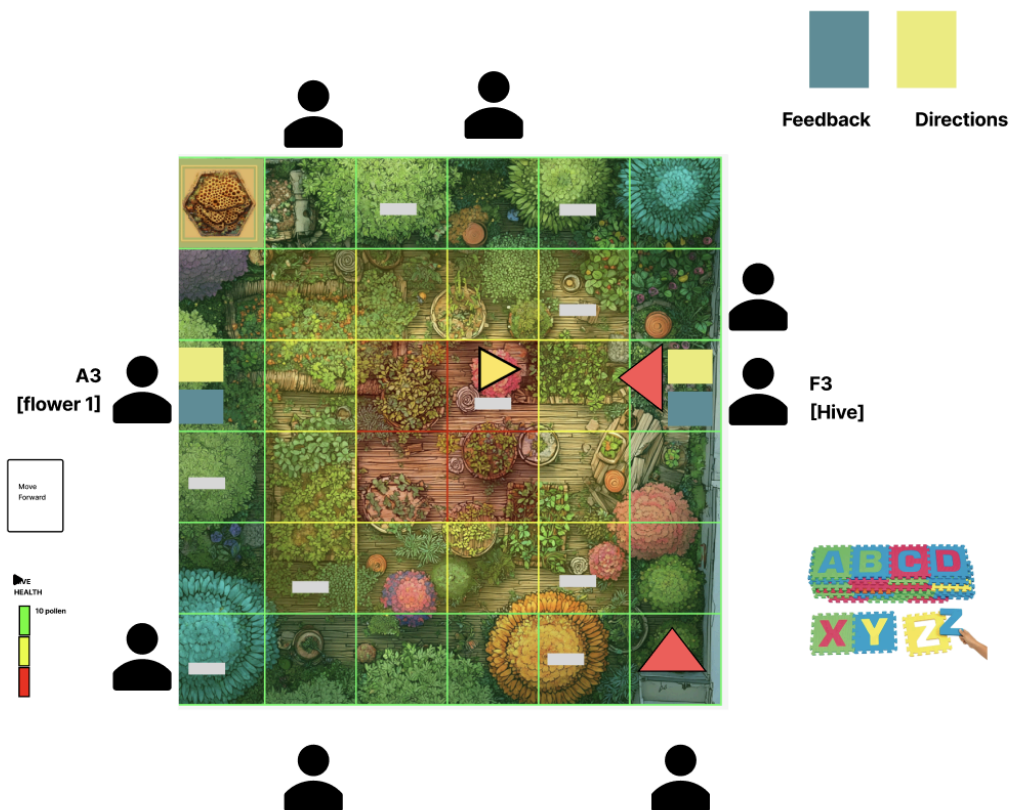
- **20 Green Tiles**
 - Few or no movement limitations
 - Pollen often low quality
 - Typically border tiles
- **12 Yellow Tiles**
 - Moderate obstacles
 - Movement restricted most of the time
 - Pollen sometimes high quality
 - Placed just inside green tiles
- **4 Red Tiles**
 - Many obstacles
 - Movement always restricted
 - Pollen almost always high quality
 - Located at the center of the board

The play mat may be **double-sided**, with a more challenging configuration on the reverse.

Hive

- Starting and return point

- Can be placed anywhere in the garden to adjust difficulty
- Serves as a safe space for reflection, dancing, and reset



Game Objective

There is **no traditional win condition**. Instead, players:

- Collaboratively collect pollen
- Navigate the garden using programmed Bee-Bots
- Return pollen to the Hive
- Maintain and improve **Hive Health**

This emphasis on **sustained collective well-being** removes individual pressure and reinforces collaboration, risk-management, and shared responsibility.

Programing and Computational Learning

Core Skills

- Sequencing
- Debugging

- Problem decomposition
- Physical computing

Grade Progression

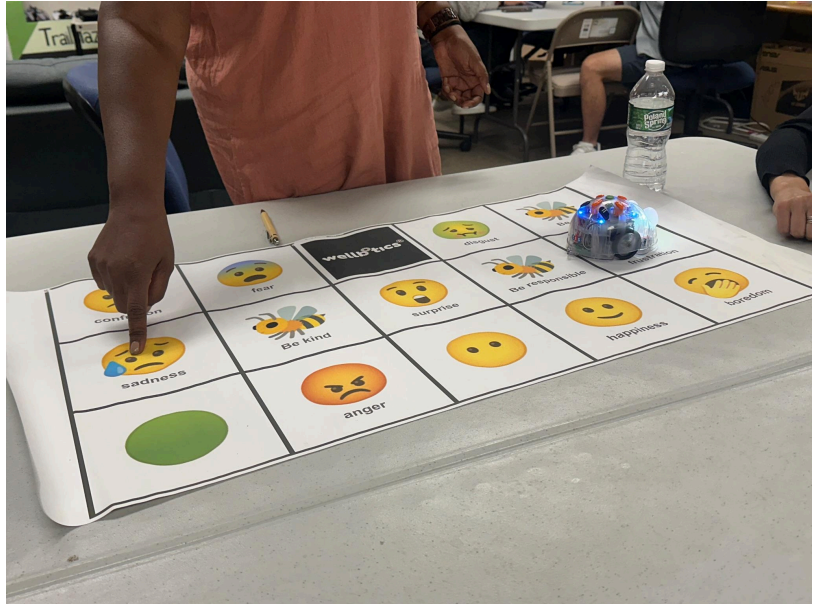
- **Kindergarten:** 1–2 step sequencing using visual direction cards
- **1st Grade:** 2–3 step chained commands
- **2nd Grade:** Introduction to simple conditionals (e.g., “if blocked, turn left”)

Game Application

Players program Bee-Bots to:

- Navigate obstacles
- Collect pollen
- Return to the Hive

Unexpected outcomes trigger **debugging**, reinforcing computational thinking through embodied trial and error.



Core System (Game Mechanics)

Each tile has a **deck of cards** (number TBD; ~10 suggested, explicitly marked for testing).

Card Categories

1. Collect Pollen Cards

- **Visuals:** Bright flowers, pollen icons
- **Action:** Attach pollen to the Bee-Bot
- **SEL Integration:**
 - “The hive feels happy when you bring pollen”
 - Reinforces contribution and social awareness

2. Obstacle / Movement Limitation Cards

- **Visuals:** Wind, vines, puddles, branches
- **Actions:**
 - Extra programming cost
 - Lose pollen
 - Move backward
 - Reprogram route
- **SEL Integration:**
 - “Frustration Station! Take a deep breath.”
 - Promotes persistence, emotional regulation, and teamwork

3. Dance Cards

- **Visuals:** Dancing bees, emotive icons
- **Actions:**
 - Perform shared dances
 - Earn Hive Health
- **SEL Integration:**
 - Emotional expression through movement
 - Mindfulness (“move slowly and notice your body”)
 - Communication (“can your friends guess how you feel?”)
 - Collaboration (“create a teamwork move together”)

Sequence of Play

1. **Queen Bee Gives Instructions**
 - Destination tile
 - Maximum number of programming cards
2. **Players Program Bee-Bots**
 - Select and sequence cards
3. **Bee-Bots Execute Programs**
 - Physical execution on the board
4. **Card Interaction**
 - Draw pollen, obstacle, or dance cards
5. **Dance / Reflection Phase (Triggered)**
 - Movement and facilitated discussion
6. **Assess Hive Health**
 - Queen visually demonstrates hive condition
7. **Reset / Collect**
 - New cards, return to hive, or continuation

Design Status & Open Questions

The prototype explicitly acknowledges unresolved design considerations:

- Card management complexity for young learners
- Tracking pollen and hive health
- Balancing freedom vs constraint
- Motivating long-distance travel
- Managing risk and fairness
- Avoiding overly punitive mechanics
- Clarifying direction language (clockwise vs counter-clockwise)

These questions mark the system as **iterative and intentionally unfinished**, appropriate for prototyping and classroom testing.

Conclusion

The Bee-Bot Garden Prototype demonstrates how **robot programming can function as an emotional mirror**, allowing young learners to externalize, test, reflect on, and revise their behaviors through play. By embedding SEL within **game mechanics, physical movement, and collaborative goals**, the system avoids treating emotional learning as abstract or forced. Rather than teaching emotions directly, the game allows learners to **experience regulation, persistence, frustration, and connection** as system states—making emotional intelligence tangible, shareable, and developmentally appropriate.

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